```
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# Generate a synthetic dataset
X, y = make_classification(n_samples=1000, n_features=20, n_informative=15, n_redundant=5, random_state=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

```
# Initialize and train Decision Tree
dt_clf = DecisionTreeClassifier(random_state=42)
dt_clf.fit(X_train, y_train)
dt_pred = dt_clf.predict(X_test)

# Initialize and train Random Forest
rf_clf = RandomForestClassifier(n_estimators=100, random_state=42)
rf_clf.fit(X_train, y_train)
rf_pred = rf_clf.predict(X_test)
```

```
# Compare Performance
dt_acc = accuracy_score(y_test, dt_pred)
rf_acc = accuracy_score(y_test, rf_pred)

print(f"Decision Tree Accuracy: {dt_acc:.4f}")
print(f"Random Forest Accuracy: {rf_acc:.4f}")

print("\nDecision Tree Report:")
print(classification_report(y_test, dt_pred))

print("\nRandom Forest Report:")
print(classification_report(y_test, rf_pred))
```

```
Decision Tree Accuracy: 0.7967
Random Forest Accuracy: 0.8867

Decision Tree Report:
              precision    recall  f1-score   support

           0       0.83      0.78      0.80       160
           1       0.76      0.82      0.79       140

    accuracy                           0.80       300
   macro avg       0.80      0.80      0.80       300
weighted avg       0.80      0.80      0.80       300

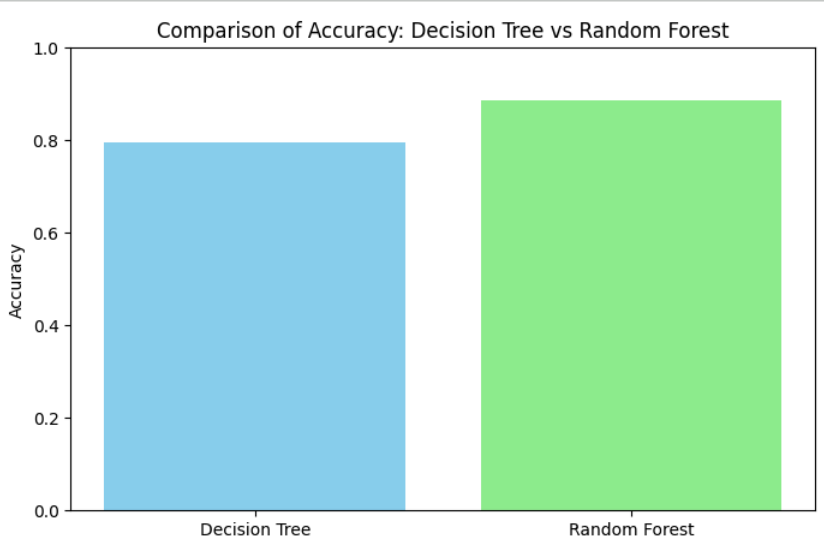

Random Forest Report:
              precision    recall  f1-score   support

           0       0.90      0.89      0.89       160
           1       0.87      0.89      0.88       140

    accuracy                           0.89       300
   macro avg       0.89      0.89      0.89       300
weighted avg       0.89      0.89      0.89       300
```

```
# Visual Comparison
models = ['Decision Tree', 'Random Forest']
accuracies = [dt_acc, rf_acc]

plt.figure(figsize=(8, 5))
plt.bar(models, accuracies, color=['skyblue', 'lightgreen'])
plt.ylabel('Accuracy')
plt.title('Comparison of Accuracy: Decision Tree vs Random Forest')
plt.ylim(0, 1.0)
plt.show()
```


Comparison of Accuracy: Decision Tree vs Random Forest
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Decision Tree
Random Forest